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Corso: CyberSecurity

Attività: Ricognizione della macchina Metasploitable con Nmap e strumenti complementari

Ricognizione della macchina Metasploitable

Ambiente e indirizzi usati

Le scansioni sono state eseguite da Kali. Metasploitable è il target principale; Windows è stato usato come target aggiuntivo per le verifiche SMB e della rete locale.

Macchina	Indirizzo	Ruolo
Kali	192.168.50.100	Macchina che esegue le scansioni
Metasploitable	192.168.51.101	Target principale
Windows	192.168.50.102	Target aggiuntivo
pfSense	192.168.50.1 / 192.168.51.1	Gateway tra le due reti

1 Scoperta dell'host Metasploitable

```
nmap -sn -PE 192.168.51.101
```

Scopo. Verificare se l'indirizzo risponde alla scoperta ICMP. -sn esegue la sola scoperta dell'host e -PE usa ICMP Echo.

Risultato. 192.168.51.101 risulta attivo.

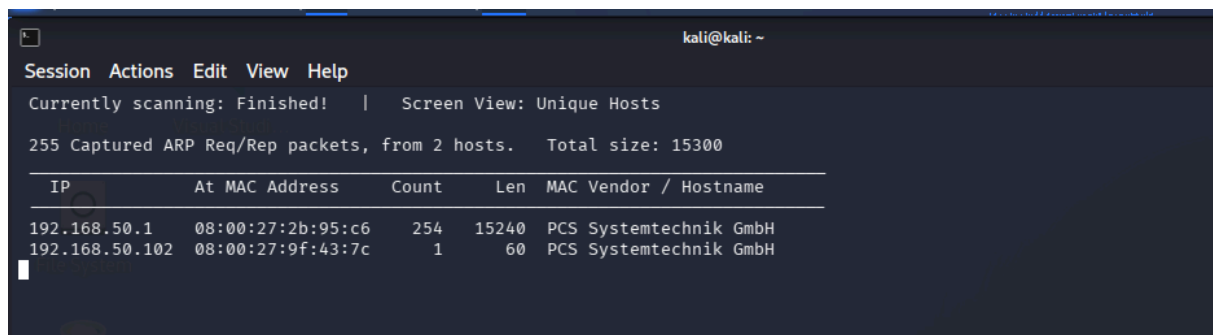
```
(kali㉿kali)-[~]  
$ nmap -sn -PE 192.168.51.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 07:30 -0400  
Nmap scan report for 192.168.51.101  
Host is up (0.0045s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.54 seconds
```

2 Scoperta dei dispositivi nella rete locale

```
sudo netdiscover -r 192.168.50.0/24
```

Scopo. Usare ARP per elencare i dispositivi direttamente presenti nella rete locale. ARP non attraversa il router.

Risultato. Sono stati rilevati pfSense (192.168.50.1) e Windows (192.168.50.102). Kali non compare perché è la macchina che esegue la scansione.



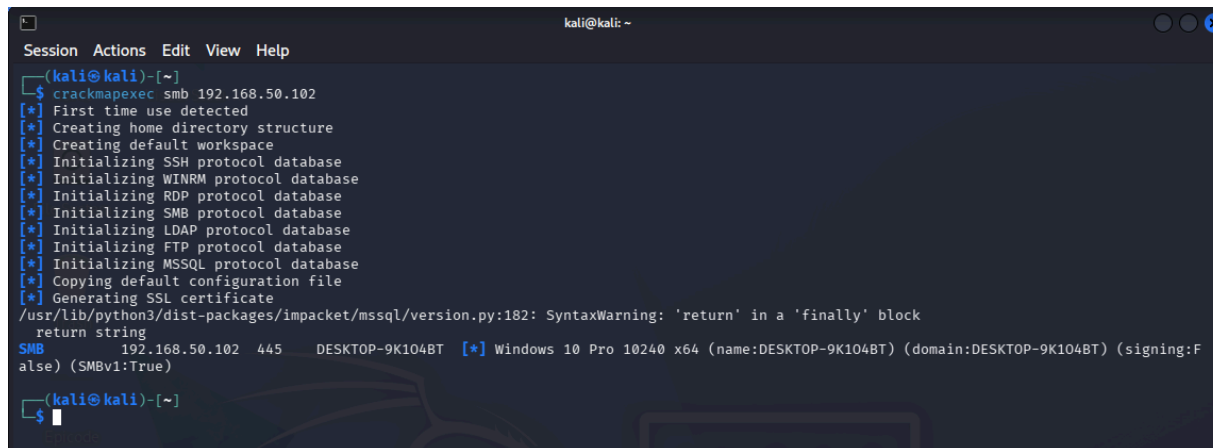
IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.50.1	08:00:27:2b:95:c6	254	15240	PCS Systemtechnik GmbH
192.168.50.102	08:00:27:9f:43:7c	1	60	PCS Systemtechnik GmbH

3 Enumerazione SMB del target Windows aggiuntivo

```
crackmapexec smb 192.168.50.102
```

Scopo. Verificare se Windows espone SMB e leggere le informazioni pubbliche sulla porta 445.

Risultato. SMB risponde sulla porta 445. L'output identifica Windows 10 Pro 10240 x64, il nome DESKTOP-9K104BT e SMBv1 attivo. Il SyntaxWarning di Impacket non ha bloccato il comando.



```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ crackmapexec smb 192.168.50.102  
[*] First time use detected  
[*] Creating home directory structure  
[*] Creating default workspace  
[*] Initializing SSH protocol database  
[*] Initializing WINRM protocol database  
[*] Initializing RDP protocol database  
[*] Initializing SMB protocol database  
[*] Initializing LDAP protocol database  
[*] Initializing FTP protocol database  
[*] Initializing MSSQL protocol database  
[*] Copying default configuration file  
[*] Generating SSL certificate  
/usr/lib/python3/dist-packages/impacket/mssql/version.py:182: SyntaxWarning: 'return' in a 'finally' block  
  return string  
SMB 192.168.50.102 445 DESKTOP-9K104BT [*] Windows 10 Pro 10240 x64 (name:DESKTOP-9K104BT) (domain:DESKTOP-9K104BT) (signing:False) (SMBv1:True)  
(kali@kali)-[~]  
$
```

4 Scansione rapida delle porte principali

```
nmap 192.168.51.101 --top-ports 10 --open
```

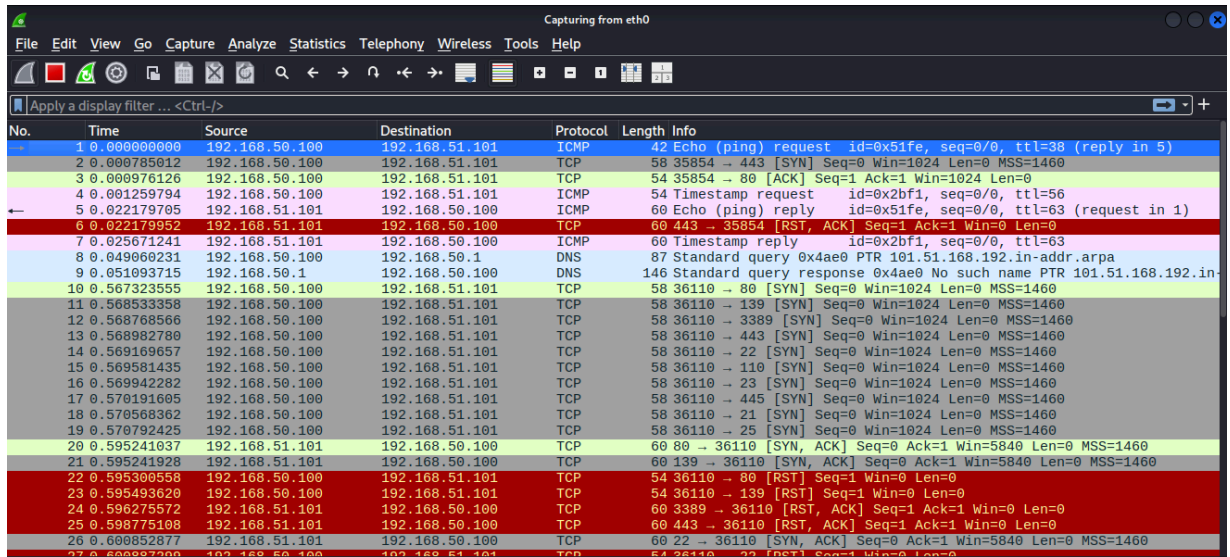
Scopo. Controllare rapidamente le dieci porte più comuni. --open mostra solo le porte aperte.

Risultato. Sono aperte le porte 21, 22, 23, 25, 80, 139 e 445; tre porte controllate risultano chiuse.

```
(kali㉿kali)-[~]  
$ nmap 192.168.51.101 --top-ports 10 --open  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 07:33 -0400  
Nmap scan report for 192.168.51.101  
Host is up (0.029s latency).  
Not shown: 3 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
80/tcp    open  http  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
  
Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds
```

Analisi Wireshark associata alla scansione

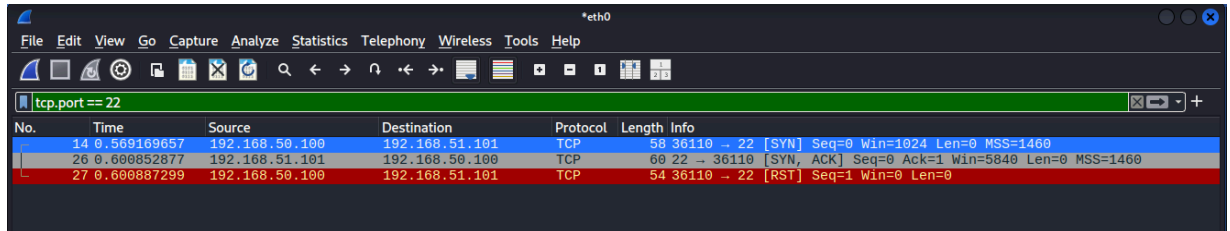
Mostra richieste ICMP, richieste TCP SYN e risposte SYN-ACK o RST. Il filtro tcp.port == 22 isola il controllo della porta SSH.



Wireshark packet capture showing various network protocols. The display filter is empty. The packet list shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.50.100	192.168.51.101	ICMP	42	Echo (ping) request id=0x51fe, seq=0/0, ttl=38 (reply in 5)
2	0.000705012	192.168.50.100	192.168.51.101	TCP	58	35854 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
3	0.000976126	192.168.50.100	192.168.51.101	TCP	54	35854 → 80 [ACK] Seq=1 Ack=1 Win=1024 Len=0
4	0.001259794	192.168.50.100	192.168.51.101	ICMP	54	Timestamp request id=0x2bf1, seq=0/0, ttl=56
5	0.022179705	192.168.51.101	192.168.50.100	ICMP	60	Echo (ping) reply id=0x51fe, seq=0/0, ttl=63 (request in 1)
6	0.022179952	192.168.51.101	192.168.50.100	TCP	60	443 → 35854 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
7	0.025671241	192.168.51.101	192.168.50.100	ICMP	60	Timestamp reply id=0x2bf1, seq=0/0, ttl=63
8	0.049060231	192.168.50.100	192.168.50.1	DNS	87	Standard query 0x4ae0 PTR 101.51.168.192.in-addr.arpa
9	0.051093715	192.168.50.1	192.168.50.100	DNS	146	Standard query response 0x4ae0 No such name PTR 101.51.168.192.in-
10	0.567323555	192.168.50.100	192.168.51.101	TCP	58	36110 → 80 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
11	0.568533358	192.168.50.100	192.168.51.101	TCP	58	36110 → 139 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
12	0.568768566	192.168.50.100	192.168.51.101	TCP	58	36110 → 3389 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
13	0.568982780	192.168.50.100	192.168.51.101	TCP	58	36110 → 443 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
14	0.569169657	192.168.50.100	192.168.51.101	TCP	58	36110 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
15	0.569581435	192.168.50.100	192.168.51.101	TCP	58	36110 → 110 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
16	0.569942282	192.168.50.100	192.168.51.101	TCP	58	36110 → 23 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
17	0.570191605	192.168.50.100	192.168.51.101	TCP	58	36110 → 445 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
18	0.570568362	192.168.50.100	192.168.51.101	TCP	58	36110 → 21 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
19	0.570792425	192.168.50.100	192.168.51.101	TCP	58	36110 → 25 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
20	0.595241037	192.168.51.101	192.168.50.100	TCP	60	80 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
21	0.595241928	192.168.51.101	192.168.50.100	TCP	60	139 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
22	0.595300558	192.168.50.100	192.168.51.101	TCP	54	36110 → 80 [RST] Seq=1 Win=0 Len=0
23	0.595493620	192.168.50.100	192.168.51.101	TCP	54	36110 → 139 [RST] Seq=1 Win=0 Len=0
24	0.596275572	192.168.51.101	192.168.50.100	TCP	60	3389 → 36110 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
25	0.598775108	192.168.51.101	192.168.50.100	TCP	60	443 → 36110 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
26	0.600852877	192.168.51.101	192.168.50.100	TCP	60	22 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
27	0.600887299	192.168.50.100	192.168.51.101	TCP	54	36110 → 22 [RST] Seq=1 Win=0 Len=0

26	0.600852877	192.168.51.101	192.168.50.100	TCP	60	22 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
27	0.600887299	192.168.50.100	192.168.51.101	TCP	54	36110 → 22 [RST] Seq=1 Win=0 Len=0
28	0.601602071	192.168.51.101	192.168.50.100	TCP	60	110 → 36110 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
29	0.602027276	192.168.51.101	192.168.50.100	TCP	60	23 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
30	0.602027419	192.168.51.101	192.168.50.100	TCP	60	445 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
31	0.602040209	192.168.50.100	192.168.51.101	TCP	54	36110 → 23 [RST] Seq=1 Win=0 Len=0
32	0.602105397	192.168.50.100	192.168.51.101	TCP	54	36110 → 445 [RST] Seq=1 Win=0 Len=0
33	0.602939464	192.168.51.101	192.168.50.100	TCP	60	21 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
34	0.602949010	192.168.50.100	192.168.51.101	TCP	54	36110 → 21 [RST] Seq=1 Win=0 Len=0
35	0.603349751	192.168.51.101	192.168.50.100	TCP	60	25 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
36	0.603357352	192.168.50.100	192.168.51.101	TCP	54	36110 → 25 [RST] Seq=1 Win=0 Len=0
37	5.178526344	PCSSystemtec_94:c1:...	PCSSystemtec_2b:95:...	ARP	42	Who has 192.168.50.1? Tell 192.168.50.100
38	5.180306317	PCSSystemtec_2b:95:...	PCSSystemtec_94:c1:...	ARP	60	192.168.50.1 is at 08:00:27:2b:95:c6



Wireshark packet capture with display filter tcp.port == 22. The packet list shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
14	0.569169657	192.168.50.100	192.168.51.101	TCP	58	36110 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
26	0.600852877	192.168.51.101	192.168.50.100	TCP	60	22 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
27	0.600887299	192.168.50.100	192.168.51.101	TCP	54	36110 → 22 [RST] Seq=1 Win=0 Len=0

*eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.port == 22

No.	Time	Source	Destination	Protocol	Length	Info
14	0.569169657	192.168.50.100	192.168.51.101	TCP	58	36110 → 22 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
26	0.600852877	192.168.51.101	192.168.50.100	TCP	60	22 → 36110 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1460
27	0.600887299	192.168.50.100	192.168.51.101	TCP	54	36110 → 22 [RST] Seq=1 Win=0 Len=0

Frame 26: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface eth0, id 0

Ethernet II, Src: PCSSystemtec_2b:95:c6 (08:00:27:2b:95:c6), Dst: PCSSystemtec_94:c1:4e (08:00:27:94:c1:4e)

Internet Protocol Version 4, Src: 192.168.51.101, Dst: 192.168.50.100

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 44

Identification: 0x0000 (0)

010. = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 63

Protocol: TCP (6)

Header Checksum: 0x54b2 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.51.101

Destination Address: 192.168.50.100

[Stream index: 0]

Transmission Control Protocol, Src Port: 22, Dst Port: 36110, Seq: 0, Ack: 1, Len: 0

Source Port: 22

Destination Port: 36110

[Stream index: 6]

[Stream Packet Number: 2]

[Conversation completeness: Incomplete (35)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 2588158973

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 1999855886

0110 = Header Length: 24 bytes (6)

Flags: 0x012 (SYN, ACK)

Window: 5840

[Calculated window size: 5840]

Checksum: 0xbe61 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (4 bytes), Maximum segment size

[Timestamp]

Source Port (tcp.srcport), 2 bytes

Packets: 38 - Displayed: 3 (7.9%) - Dropped: 0 (0.0%) | Profile: Default

5 Scansione completa TCP con rilevamento dei servizi

```
sudo nmap -Pn -p- -sV --reason 192.168.51.101
```

Scopo. Controllare tutte le 65.535 porte (-p-), saltare la fase di ping (-Pn), identificare i servizi (-sV) e mostrare il motivo della classificazione (--reason). La traccia riportava anche --dns-server ns, ma ns non era un server DNS concreto.

Risultato. Sono state rilevate 65.505 porte chiuse e numerose porte aperte. Le porte aperte riportano il motivo syn-ack e TTL 63.

```
(kali@kali)-[~]
$ sudo nmap -Pn -p- -sV --reason 192.168.51.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 08:53 -0400
Nmap scan report for 192.168.51.101
Host is up, received user-set (0.076s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      REASON      VERSION
21/tcp    open  ftp          syn-ack ttl 63 vsftpd 2.3.4
22/tcp    open  ssh          syn-ack ttl 63 OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       syn-ack ttl 63 Linux telnetd
25/tcp    open  smtp         syn-ack ttl 63 Postfix smtpd
53/tcp    open  domain       syn-ack ttl 63 ISC BIND 9.4.2
80/tcp    open  http         syn-ack ttl 63 Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      syn-ack ttl 63 2 (RPC #100000)
139/tcp   open  netbios-ssn  syn-ack ttl 63 Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  syn-ack ttl 63 Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         syn-ack ttl 63 netkit-rsh rexecd
513/tcp   open  login?       syn-ack ttl 63
514/tcp   open  tcpwrapped   syn-ack ttl 63
1099/tcp  open  java-rmi     syn-ack ttl 63 GNU Classpath grmiregistry
1524/tcp  open  bindshell    syn-ack ttl 63 Metasploitable root shell
2049/tcp  open  nfs          syn-ack ttl 63 2-4 (RPC #100003)
2121/tcp  open  ftp          syn-ack ttl 63 ProFTPD 1.3.1
3306/tcp  open  mysql        syn-ack ttl 63 MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      syn-ack ttl 63 distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   syn-ack ttl 63 PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          syn-ack ttl 63 VNC (protocol 3.3)
6000/tcp  open  X11          syn-ack ttl 63 (access denied)
6667/tcp  open  irc          syn-ack ttl 63 UnrealIRCd
6697/tcp  open  irc          syn-ack ttl 63 UnrealIRCd
8009/tcp  open  ajp13        syn-ack ttl 63 Apache Jserv (Protocol v1.3)
8180/tcp  open  http         syn-ack ttl 63 Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb          syn-ack ttl 63 Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbl)
36926/tcp open  mountd       syn-ack ttl 63 1-3 (RPC #100005)
40917/tcp open  java-rmi     syn-ack ttl 63 GNU Classpath grmiregistry
41803/tcp open  status       syn-ack ttl 63 1 (RPC #100024)
51534/tcp open  nlockmgr     syn-ack ttl 63 1-4 (RPC #100021)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 264.02 seconds
```

6 Scansione TCP e UDP con Unicornscan

```
sudo us -mT -Iv 192.168.51.101:a -r 3000 -R 3 && sudo us -mU -Iv 192.168.51.101:a -r 3000 -R 3
```

Scopo. Confrontare un motore alternativo. -mT usa TCP, -mU UDP, -Iv aumenta il dettaglio, :a indica tutte le porte, -r 3000 il ritmo e -R 3 i ritentativi.

Risultato. La parte TCP rileva varie porte aperte. Per UDP l'output visibile riporta 0 pacchetti ricevuti: indica assenza di risposte osservate, non la prova che ogni porta UDP sia chiusa.

```
(kali@kali)-[~]
$ sudo us -mT -Iv 192.168.51.101:a -r 3000 -R 3 && sudo us -mU -Iv 192.168.51.101:a -r 3000 -R 3
adding 192.168.51.101/32 mode 'TCPscan' ports 'a' pps 3000
using interface(s) eth0
scanning 1.00e+00 total hosts with 1.97e+05 total packets, should take a little longer than 1 Minutes, 12 Seconds
TCP open 192.168.51.101:22 ttl 63
TCP open 192.168.51.101:512 ttl 63
TCP open 192.168.51.101:53 ttl 63
TCP open 192.168.51.101:36926 ttl 63
TCP open 192.168.51.101:5432 ttl 63
TCP open 192.168.51.101:2121 ttl 63
TCP open 192.168.51.101:139 ttl 63
TCP open 192.168.51.101:6000 ttl 63
TCP open 192.168.51.101:41803 ttl 63
TCP open 192.168.51.101:514 ttl 63
TCP open 192.168.51.101:80 ttl 63
TCP open 192.168.51.101:8180 ttl 63
TCP open 192.168.51.101:8009 ttl 63
TCP open 192.168.51.101:513 ttl 63
TCP open 192.168.51.101:3632 ttl 63
TCP open 192.168.51.101:6667 ttl 63
TCP open 192.168.51.101:21 ttl 63
TCP open 192.168.51.101:51534 ttl 63
TCP open 192.168.51.101:8787 ttl 63
sender statistics 2691.0 pps with 196608 packets sent total
listener statistics 48049 packets recieved 0 packets dropped and 0 interface drops
TCP open ftp[ 21] from 192.168.51.101 ttl 63
TCP open ssh[ 22] from 192.168.51.101 ttl 63
TCP open domain[ 53] from 192.168.51.101 ttl 63
TCP open http[ 80] from 192.168.51.101 ttl 63
TCP open netbios-ssn[ 139] from 192.168.51.101 ttl 63
TCP open exec[ 512] from 192.168.51.101 ttl 63
TCP open login[ 513] from 192.168.51.101 ttl 63
TCP open shell[ 514] from 192.168.51.101 ttl 63
TCP open scientia-ssdb[ 2121] from 192.168.51.101 ttl 63
TCP open distcc[ 3632] from 192.168.51.101 ttl 63
TCP open postgresql[ 5432] from 192.168.51.101 ttl 63
TCP open x11[ 6000] from 192.168.51.101 ttl 63
TCP open irc[ 6667] from 192.168.51.101 ttl 63
TCP open unknown[ 8009] from 192.168.51.101 ttl 63
TCP open unknown[ 8180] from 192.168.51.101 ttl 63
TCP open msgsrvr[ 8787] from 192.168.51.101 ttl 63
TCP open unknown[36926] from 192.168.51.101 ttl 63
TCP open unknown[41803] from 192.168.51.101 ttl 63
TCP open unknown[51534] from 192.168.51.101 ttl 63
adding 192.168.51.101/32 mode 'UDPscan' ports 'a' pps 3000
using interface(s) eth0
scanning 1.00e+00 total hosts with 1.97e+05 total packets, should take a little longer than 1 Minutes, 12 Seconds
sender statistics 2949.6 pps with 196635 packets sent total
listener statistics 0 packets recieved 0 packets dropped and 0 interface drops
```

7 Scansione SYN con rilevamento dei servizi

```
sudo nmap -sS -sV -T4 192.168.51.101
```

Scopo. Eseguire una scansione SYN (-sS), identificare i servizi (-sV) e usare una temporizzazione più rapida (-T4).

Risultato. Il risultato conferma le porte e le versioni osservate con Nmap; il target risulta attivo.

```
(kali@kali)~[~]
$ sudo nmap -sS -sV -T4 192.168.51.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 09:17 -0400
Nmap scan report for 192.168.51.101
Host is up (0.42s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
21/tcp    open  ftp              vsftpd 2.3.4
22/tcp    open  ssh              OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet           Linux telnetd
25/tcp    open  smtp             Postfix smtpd
53/tcp    open  domain          ISC BIND 9.4.2
80/tcp    open  http             Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind          2 (RPC #100000)
139/tcp   open  netbios-ssn     Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn     Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec            netkit-rsh rshcd
513/tcp   open  login           OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi        GNU Classpath grmiregistry
1524/tcp  open  bindshell       Metasploitable root shell
2049/tcp  open  nfs             2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql           MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql      PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc             VNC (protocol 3.3)
6000/tcp  open  X11             (access denied)
6667/tcp  open  irc             UnrealIRCd
8009/tcp  open  ajp13           Apache Jserv (Protocol v1.3)
8180/tcp  open  http            Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.80 seconds
```

8 Scansione delle porte con Hping3

```
sudo hping3 --scan known 192.168.51.101
```

Scopo. Inviare sonde TCP verso un insieme di porte note e confrontare il comportamento con Nmap.

Risultato. La scansione termina, ma non presenta righe di porte aperte e riporta molte porte come non rispondenti. Questo non contraddice Nmap: Hping3 non ha ricevuto risposte utilizzabili alle proprie sonde.

```

[~]$ sudo htop --scan kernel 192.168.21.101
Scanning 192.168.21.101 (SIO-SMB-1-101), port 80
260 ports to scan, use -v to see all the replies
[~]$
[~] server name | flags | ttl | ul | win | len |
-----+-----+---+---+---+---+
All replies received. Done.
Not responding ports: (1 tcpmux) (2 nmap) (4 echo) (6 zip) (7 echo) (9 discard) (11 systat) (13 daytime) (15 netstat) (17 qotd) (19 chargen) (20 ftp-data) (21 ftp) (22 ssh) (23 telnet) (25 smtp) (37 time) (44 whois) (49 tacacs) (53 domain) (67 bootps) (68 bootpc) (69 frp) (70 goopher) (79 finger) (80 http) (88 kerberos) (142 iso-tcp) (146 acis) (147 ntp) (148 pop3) (149 pop3s) (140 ntp) (151 nmap) (152 ntp) (153 smtp) (157 netbios-ss) (158 netbios-ss) (144 ldap) (161 snmp) (164 smptpr) (165 cnc) (166) (167) (168) (169) (170) (171) (172) (173) (174) (175) (176) (177) (178) (179) (180) (181) (182) (183) (184) (185) (186) (187) (188) (189) (190) (191) (192) (193) (194) (195) (196) (197) (198) (199) (200) (201) (202) (203) (204) (205) (206) (207) (208) (209) (210) (211) (212) (213) (214) (215) (216) (217) (218) (219) (220) (221) (222) (223) (224) (225) (226) (227) (228) (229) (230) (231) (232) (233) (234) (235) (236) (237) (238) (239) (240) (241) (242) (243) (244) (245) (246) (247) (248) (249) (250) (251) (252) (253) (254) (255) (256) (257) (258) (259) (260) (261) (262) (263) (264) (265) (266) (267) (268) (269) (270) (271) (272) (273) (274) (275) (276) (277) (278) (279) (280) (281) (282) (283) (284) (285) (286) (287) (288) (289) (290) (291) (292) (293) (294) (295) (296) (297) (298) (299) (300) (301) (302) (303) (304) (305) (306) (307) (308) (309) (310) (311) (312) (313) (314) (315) (316) (317) (318) (319) (320) (321) (322) (323) (324) (325) (326) (327) (328) (329) (330) (331) (332) (333) (334) (335) (336) (337) (338) (339) (340) (341) (342) (343) (344) (345) (346) (347) (348) (349) (350) (351) (352) (353) (354) (355) (356) (357) (358) (359) (360) (361) (362) (363) (364) (365) (366) (367) (368) (369) (370) (371) (372) (373) (374) (375) (376) (377) (378) (379) (380) (381) (382) (383) (384) (385) (386) (387) (388) (389) (390) (391) (392) (393) (394) (395) (396) (397) (398) (399) (400) (401) (402) (403) (404) (405) (406) (407) (408) (409) (410) (411) (412) (413) (414) (415) (416) (417) (418) (419) (420) (421) (422) (423) (424) (425) (426) (427) (428) (429) (430) (431) (432) (433) (434) (435) (436) (437) (438) (439) (440) (441) (442) (443) (444) (445) (446) (447) (448) (449) (450) (451) (452) (453) (454) (455) (456) (457) (458) (459) (460) (461) (462) (463) (464) (465) (466) (467) (468) (469) (470) (471) (472) (473) (474) (475) (476) (477) (478) (479) (480) (481) (482) (483) (484) (485) (486) (487) (488) (489) (490) (491) (492) (493) (494) (495) (496) (497) (498) (499) (500) (501) (502) (503) (504) (505) (506) (507) (508) (509) (510) (511) (512) (513) (514) (515) (516) (517) (518) (519) (520) (521) (522) (523) (524) (525) (526) (527) (528) (529) (530) (531) (532) (533) (534) (535) (536) (537) (538) (539) (540) (541) (542) (543) (544) (545) (546) (547) (548) (549) (550) (551) (552) (553) (554) (555) (556) (557) (558) (559) (560) (561) (562) (563) (564) (565) (566) (567) (568) (569) (570) (571) (572) (573) (574) (575) (576) (577) (578) (579) (580) (581) (582) (583) (584) (585) (586) (587) (588) (589) (590) (591) (592) (593) (594) (595) (596) (597) (598) (599) (600) (601) (602) (603) (604) (605) (606) (607) (608) (609) (610) (611) (612) (613) (614) (615) (616) (617) (618) (619) (620) (621) (622) (623) (624) (625) (626) (627) (628) (629) (630) (631) (632) (633) (634) (635) (636) (637) (638) (639) (640) (641) (642) (643) (644) (645) (646) (647) (648) (649) (650) (651) (652) (653) (654) (655) (656) (657) (658) (659) (660) (661) (662) (663) (664) (665) (666) (667) (668) (669) (670) (671) (672) (673) (674) (675) (676) (677) (678) (679) (680) (681) (682) (683) (684) (685) (686) (687) (688) (689) (690) (691) (692) (693) (694) (695) (696) (697) (698) (699) (700) (701) (702) (703) (704) (705) (706) (707) (708) (709) (710) (711) (712) (713) (714) (715) (716) (717) (718) (719) (720) (721) (722) (723) (724) (725) (726) (727) (728) (729) (730) (731) (732) (733) (734) (735) (736) (737) (738) (739) (740) (741) (742) (743) (744) (745) (746) (747) (748) (749) (750) (751) (752) (753) (754) (755) (756) (757) (758) (759) (760) (761) (762) (763) (764) (765) (766) (767) (768) (769) (770) (771) (772) (773) (774) (775) (776) (777) (778) (779) (780) (781) (782) (783) (784) (785) (786) (787) (788) (789) (790) (791) (792) (793) (794) (795) (796) (797) (798) (799) (800) (801) (802) (803) (804) (805) (806) (807) (808) (809) (810) (811) (812) (813) (814) (815) (816) (817) (818) (819) (820) (821) (822) (823) (824) (825) (826) (827) (828) (829) (830) (831) (832) (833) (834) (835) (836) (837) (838) (839) (840) (841) (842) (843) (844) (845) (846) (847) (848) (849) (850) (851) (852) (853) (854) (855) (856) (857) (858) (859) (860) (861) (862) (863) (864) (865) (866) (867) (868) (869) (870) (871) (872) (873) (874) (875) (876) (877) (878) (879) (880) (881) (882) (883) (884) (885) (886) (887) (888) (889) (890) (891) (892) (893) (894) (895) (896) (897) (898) (899) (900) (901) (902) (903) (904) (905) (906) (907) (908) (90
```

9 Verifica aggiuntiva Hping3 su Windows

```
sudo hping3 --scan known 192.168.50.102
```

Scopo. Ripetere il controllo Hping3 su Windows per osservare un secondo sistema del laboratorio.

Risultato. Anche qui l'output riporta porte non rispondenti senza fornire un inventario completo dei servizi.

```
(kali@kali)-[~]
$ sudo hping3 --scan known 192.168.50.102
Scanning 192.168.50.102 (192.168.50.102), port known
260 ports to scan, use -V to see all the replies
+-----+-----+-----+-----+-----+
|port| serv name | flags |ttl| id | win | len |
+-----+-----+-----+-----+-----+
All replies received. Done.
Not responding ports: (1 tcpmux) (2 nbp) (4 echo) (6 zip) (7 echo) (9 discard) (11 systat) (13 daytime) (15 netstat) (17 qotd) (19 chargen)
(20 ftp-data) (21 ftp) (22 ssh) (23 telnet) (25 smtp) (37 time) (43 whois) (49 tacacs) (53 domain) (67 bootps) (68 bootpc) (69 tftp) (70 gopher) (79 finger) (80 http) (88 kerberos) (102 iso-tsap) (104 acr-nema) (106 poppassd) (110 pop3) (111 sunrpc) (113 auth) (119 nntp) (123 ntp) (135 epmap) (137 netbios-ns) (138 netbios-dgm) (139 netbios-ssn) (143 imap) (161 snmp) (162 snmptrap) (163 cmip-man) (164 cmip-agent) (174 mailq) (177 xdmcp) (179 bgp) (199 smux) (209 qmtp) (210 z39-50) (213 ipx) (319 ptp-event) (320 ptp-general) (345 pawsserv) (346 zserv) (369 rpc2portmap) (370 codaauth2) (371 clearcase) (389 ldap) (427 svrloc) (443 https) (444 snpp) (445 microsoft-d) (464 kpasswd) (465 submi) (487 saft) (500 isakmp) (512 exec) (513 login) (514 shell) (515 printer) (517 talk) (518 ntalk) (520 route) (538 gdomap) (540 uucp) (543 klogin) (544 kshell) (546 dhcpv6-clie) (547 dhcpv6-serv) (548 afpovertcp) (554 rtsp) (563 nntps) (587 submission) (607 nqs) (623 asf-rmcp) (628 qmqp) (631 ipp) (636 ldaps) (646 ldp) (655 tinc) (706 silc) (749 kerberos-ad) (750 kerberos4) (751 kerberos-ma) (752 passwd-serv) (754 krb-prop) (775 moira-db) (777 moira-updat) (779 moira-ureg) (783 spamd) (853 domain-s) (871 supfilesrv) (873 rsync) (989 ftps-data) (990 ftps) (992 telnets) (993 imaps) (995 pop3s) (1080 socks) (1093 proofd) (1094 rootd) (1099 rmiregistry) (1127 supfiledbg) (1178 skkserv) (1194 openvpn) (1210 predict) (1313 xtel) (1314 xtelw) (1352 lotusnote) (1433 ms-sql-s) (1434 ms-sql-m) (1524 ingreslock) (1645 sightline) (1646 sa-msg-port) (1649 kermit) (1677 groupwise) (1701 l2f) (1812 radius) (1813 radius-acct) (2000 cisco-sccp) (2049 nfs) (2086 gnutel) (2101 rtcn-sci04) (2102 zephyr-srv) (2103 zephyr-clt) (2104 zephyr-hm) (2119 gsgatekeep) (2121 iprop) (2135 gris) (2401 cvspserver) (2430 venus) (2431 venus-se) (2432 codasrv) (2433 codasrv-se) (2583 mon) (2600 zebrasrv) (2601 zebra) (2602 ripd) (2603 ripngd) (2604 ospfd) (2605 bgpd) (2606 ospf6d) (2607 ospfapi) (2608 isisd) (2628 dict) (2792 f5-globals) (2811 gsift) (2947 gpsd) (3050 gds-db) (3130 icpv2) (3205 isns) (3260 iscsi-targe) (3306 mysql) (3389 ms-wbt-serv) (3493 nut) (3632 distcc) (3689 daap) (3690 svn) (4031 suucp) (4094 sysrqd) (4190 sieve) (4353 f5-iquery) (4369 epmd) (4373 remctl) (4460 ntske) (4500 ipsec-nat-t) (4557 fax) (4559 hylafax) (4569 iax) (4691 mtn) (4899 radmin-port) (4949 munin) (5060 sip) (5061 sips) (5222 xmpp-client) (5269 xmpp-server) (5308 cfengine) (5353 mdns) (5432 postgresql) (5555 rpl) (5556 freeciv) (5666 nrpe) (5667 nsca) (5671 amqps) (5672 amqp) (5680 canna) (5683 coap) (5684 coaps) (6000 x11) (6001 x11-1) (6002 x11-2) (6003 x11-3) (6004 x11-4) (6005 x11-5) (6006 x11-6) (6007 x11-7) (6346 gnutella-sv) (6347 gnutella-rt) (6379 redis) (6444 sge-qmaster) (6445 sge-execd) (6446 mysql-proxy) (6514 syslog-tls) (6566 sane-port) (6667 ircd) (6696 babel) (6697 ircs-u) (7000 bbs) (7001 afs3-callba) (7002 afs3-prserv) (7003 afs3-vlserv) (7004 afs3-kaserv) (7005 afs3-volser) (7007 afs3-bos) (7008 afs3-update) (7009 afs3-rmtsys) (7100 fo nt-servic) (8021 zope-ftp) (8080 http-alt) (8081 tproxy) (8088 omniorb) (8140 puppet) (8990 clc-build-d) (9098 xinetd) (9101 bacula-dir) (9102 bacula-fd) (9103 bacula-sd) (9418 git) (9667 xmms2) (9673 zope) (10000 webmin) (10050 zabbix-agen) (10051 zabbix-trap) (10080 amanda) (10081 kamanda) (10082 amandaidx) (10083 amidxtape) (10809 nbd) (11112 dicom) (11371 hkp) (17500 db-lsp) (22125 dcap) (22128 gsidcap) (22273 wnn6) (24554 blink) (30865 csvnc2) (57000 dircproxy) (60177 tfido) (60179 fido)
```

10 Controllo delle porte da 1 a 1024 con Netcat

```
nc -nvz 192.168.51.101 1-1024
```

Scopo. Provare le porte TCP da 1 a 1024 senza aprire una sessione applicativa. -n evita DNS, -v mostra i dettagli e -z esegue solo il controllo.

Risultato. Sono aperte 21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513 e 514. Le porte chiuse o filtrate non vengono elencate.

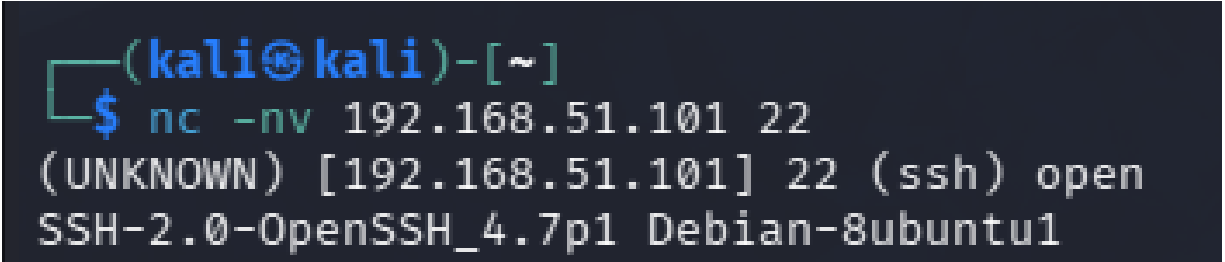
```
Session Actions Edit View Help
(kali@kali)-[~]
$ nc -nvz 192.168.51.101 1-1024
(UNKNOWN) [192.168.51.101] 514 (shell) open
(UNKNOWN) [192.168.51.101] 513 (login) open
(UNKNOWN) [192.168.51.101] 512 (exec) open
(UNKNOWN) [192.168.51.101] 445 (microsoft-ds) open
(UNKNOWN) [192.168.51.101] 139 (netbios-ssn) open
(UNKNOWN) [192.168.51.101] 111 (sunrpc) open
(UNKNOWN) [192.168.51.101] 80 (http) open
(UNKNOWN) [192.168.51.101] 53 (domain) open
(UNKNOWN) [192.168.51.101] 25 (smtp) open
(UNKNOWN) [192.168.51.101] 23 (telnet) open
(UNKNOWN) [192.168.51.101] 22 (ssh) open
(UNKNOWN) [192.168.51.101] 21 (ftp) open
```

11 Banner grabbing del servizio SSH

```
nc -nv 192.168.51.101 22
```

Scopo. Aprire una connessione TCP verso la porta 22 e leggere il banner iniziale SSH.

Risultato. Il servizio risponde con OpenSSH_4.7p1 Debian-8ubuntu1. Il banner identifica il software dichiarato, ma da solo non dimostra una vulnerabilità.



```
(kali㉿kali)-[~]  
$ nc -nv 192.168.51.101 22  
(UNKNOWN) [192.168.51.101] 22 (ssh) open  
SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1
```

12 Rilevamento delle versioni dei servizi

```
nmap -sV 192.168.51.101
```

Scopo. Inviare sonde applicative per identificare servizio e versione.

Risultato. Nmap identifica, tra gli altri, vsftpd 2.3.4, OpenSSH 4.7p1, Apache 2.2.8, BIND 9.4.2, Samba, MySQL 5.0.51a, PostgreSQL 8.3.0-8.3.7 e Apache Tomcat.

```
(kali@kali)-[~]
$ nmap -sV 192.168.51.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 08:24 -0400
Nmap scan report for 192.168.51.101
Host is up (0.30s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.42 seconds
```

13 Stima del sistema operativo

```
sudo nmap -O 192.168.51.101
```

Scopo. Stimare il sistema operativo dal comportamento dello stack TCP/IP. -O produce una stima, non legge il sistema dall'interno.

Risultato. Nmap indica un dispositivo general purpose con Linux 2.6.X e distanza di rete di due hop.

```
(kali@kali)-[~]
$ sudo nmap -O 192.168.51.101
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 08:28 -0400
Nmap scan report for 192.168.51.101
Host is up (0.16s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.15 - 2.6.26 (likely embedded), Linux 2.6.20 - 2.6.24
Network Distance: 2 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.02 seconds
```

14 Esportazione dei risultati Nmap in XML

```
sudo nmap -sV -oX metasploitable.xml 192.168.51.101
```

Scopo. Salvare i risultati Nmap in XML. -oX crea un file strutturato per altri strumenti.

Risultato. La scansione termina e il file metasploitable.xml viene creato nella cartella corrente.

```
(kali@kali)-[~]
└─$ sudo nmap -sV -oX metasploitable.xml 192.168.51.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 09:24 -0400
Nmap scan report for 192.168.51.101
Host is up (1.2s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet        Linux telnetd
25/tcp    open  smtp          Postfix smtpd
53/tcp    open  domain        ISC BIND 9.4.2
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind       2 (RPC #100000)
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec          netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi      GNU Classpath grmiregistry
1524/tcp  open  bindshell     Metasploitable root shell
2049/tcp  open  nfs           2-4 (RPC #100003)
2121/tcp  open  ftp           ProFTPD 1.3.1
3306/tcp  open  mysql         MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  X11           (access denied)
6667/tcp  open  irc           UnrealIRCd
8009/tcp  open  ajp13         Apache Jserv (Protocol v1.3)
8180/tcp  open  http          Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.47 seconds
```

```
(kali@kali)-[~]
└─$ ls -lh metasploitable.xml
-rw-r--r-- 1 root root 14K Sep 19 09:25 metasploitable.xml
```

15 Importazione dei risultati in Metasploit

```
msfconsole
db import metasploitable.xml
```

Scopo. Importare nel database locale di Metasploit i risultati già raccolti. Il database consente di interrogare host e servizi senza rileggere il file XML.

Risultato. Metasploit importa correttamente 192.168.51.101.

[illegible]

16 Test di frammentazione dei pacchetti

```
sudo nmap -f --mtu=512 192.168.51.101
```

Scopo. Ripetere la scansione usando parametri di frammentazione IP e osservare la risposta del percorso di rete.

Risultato. Nmap completa la scansione e rileva le principali porte già osservate. Il risultato dimostra che la scansione è riuscita, non che un firewall sia stato bypassato.

```
(kali㉿kali)-[~]  
$ sudo nmap -f --mtu=512 192.168.51.101  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 09:47 -0400  
Nmap scan report for 192.168.51.101  
Host is up (0.052s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 3.51 seconds
```

17 Scansione HTTP con Masscan

```
sudo masscan 192.168.51.0/24 -p80 --banner --source-ip 192.168.50.100 --rate 100
```

Scopo. Controllare rapidamente la porta 80 sulla rete 192.168.51.0/24. --banner prova a leggere il banner HTTP, --source-ip indica l'indirizzo reale di Kali e --rate 100 limita la velocità.

Risultato. Masscan rileva la porta 80 aperta su 192.168.51.101 e su 192.168.51.1. Nell'output non compare un banner HTTP leggibile.

```
(kali㉿kali)-[~]
$ sudo masscan 192.168.51.0/24 -p80 --banner --source-ip 192.168.50.100 --rate 100
Starting masscan 1.3.2 (http://bit.ly/14GZzcT) at 2026-09-19 13:49:10 GMT
Initiating SYN Stealth Scan
Scanning 256 hosts [1 port/host]
Discovered open port 80/tcp on 192.168.51.101
Discovered open port 80/tcp on 192.168.51.1
```

Riepilogo finale

La ricognizione ha confermato che Metasploitable è raggiungibile da Kali e presenta una superficie di rete ampia. Nmap con rilevamento delle versioni fornisce il quadro più completo; Netcat e Hping3 mostrano invece solo le risposte osservate dai rispettivi tipi di sonda.

Wireshark documenta il traffico reale: la sequenza SYN, SYN-ACK e RST sulla porta 22 dimostra che SSH risponde e che il controllo di Nmap chiude poi il tentativo. L'importazione XML completa il flusso Nmap -> Meta, organizzando i dati nel database locale di Kali.

Servizi principali osservati su Metasploitable

Porta	Servizio o versione	Fonte
21/tcp	FTP - vsftpd 2.3.4	Nmap -sV
22/tcp	SSH - OpenSSH 4.7p1 Debian 8ubuntu1	Nmap -sV / Netcat
23/tcp	Telnet - Linux telnetd	Nmap -sV
25/tcp	SMTP - Postfix smtpd	Nmap -sV
53/tcp	DNS - ISC BIND 9.4.2	Nmap -sV
80/tcp	HTTP - Apache httpd 2.2.8	Nmap -sV
139/tcp	NetBIOS-SSN - Samba smbd 3.X-4.X	Nmap -sV
445/tcp	Microsoft-DS - Samba smbd 3.X-4.X	Nmap -sV
1524/tcp	Metasploitable root shell - bindshell	Nmap -sV
2121/tcp	FTP - ProFTPD 1.3.1	Nmap -sV
3306/tcp	MySQL 5.0.51a-3ubuntu5	Nmap -sV
5432/tcp	PostgreSQL 8.3.0-8.3.7	Nmap -sV

Porta	Servizio o versione	Fonte
5900/tcp	VNC protocol 3.3	Nmap -sV
6667/tcp	IRC - UnrealIRCd	Nmap -sV
8009/tcp	AJP13 - Apache Jserv protocol v1.3	Nmap -sV
8180/tcp	HTTP - Apache Tomcat/Coyote JSP engine 1.1	Nmap -sV